Applied Calculus IV

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- Some examples and content in this PPT are adapted from the reference textbook (Differential Equation by Dennis G. Zill). Full credit for the original textbook material belongs to its respective author and publisher.

Week 9

Review: Chapter 1 to 4

Contents

Midterm Exam Notification

Dear Students,

This is a reminder that the midterm exam will take place on November 30.

Please note the exam scope clearly:

> The exam covers Chapters 1, 2, and 4 of the textbook. For Chapter 4, only the subsections we covered in class—4.1.1, 4.2, 4.3, 4.4, and 4.6—will be included.
>
> Additionally, all content in the lecture slides from week 1 to week 10, including exercises and examples, is also part of the exam, so please review these materials carefully.

We recommend you organize your study plan in advance to cover both the textbook chapters and slides. If you need any help, you are very welcome to my office (Mon & Wed : 14:00 to 18:00)

Best Regards

Exercises

- Determine the order of the following differential equations and state whether each equation is linear

(1) $\frac{dy}{dx} = 4x^2 - y$;

(2) $\frac{d^2y}{dx^2} - \left(\frac{dy}{dx}\right)^2 + 12xy = 0$;

(3) $\left(\frac{dy}{dx}\right)^2 + x\frac{dy}{dx} - 3y^2 = 0$;

Solution:

(1) $\frac{dy}{dx} = 4x^2 - y$;

(2) $\frac{d^2y}{dx^2} - \left(\frac{dy}{dx}\right)^2 + 12xy = 0$;

(3) $\left(\frac{dy}{dx}\right)^2 + x\frac{dy}{dx} - 3y^2 = 0$;

(1) **Order**: First-order. **Linearity**: Linear.

(2) **Order**: Second-order. **Linearity**: Nonlinear (due to $(\frac{dy}{dx})^2$).

(3) **Order**: First-order. **Linearity**: Nonlinear (due to $(\frac{dy}{dx})^2$ and y^2).

(4) $x\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 3xy = \sin x$

(5) $\frac{dy}{dx} + \cos y + 2x = 0$

(6) $\sin\left(\frac{d^2y}{dx^2}\right) + e^y = x$

Solution:

(4) $x\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 3xy = \sin x$

(5) $\frac{dy}{dx} + \cos y + 2x = 0$

(6) $\sin\left(\frac{d^2y}{dx^2}\right) + e^y = x$

(4) **Order**: Second-order; **Linearity**: Linear

(5) **Order**: First-order; **Linearity**: Nonlinear

(6) **Order**: Second-order; **Linearity**: Nonlinear

- Verify that the following functions are all solutions of the equation $\frac{d^2y}{dx^2} + \omega^2 y = 0$ where $\omega > 0$ is a constant:

1. $y = \cos \omega x$
2. $y = c_1 \cos \omega x$ (c_1 is an arbitrary constant)
3. $y = \sin \omega x$
4. $y = c_2 \sin \omega x$ (c_2 is an arbitrary constant)
5. $y = c_1 \cos \omega x + c_2 \sin \omega x$ (c_1, c_2 are arbitrary constants)
6. $y = A \sin(\omega x + B)$ (A, B are arbitrary constants)

Solution:

1. For $y = \cos \omega x$:

- $y' = -\omega \sin \omega x$
- $y'' = -\omega^2 \cos \omega x$
- $y'' + \omega^2 y = -\omega^2 \cos \omega x + \omega^2 \cos \omega x = 0$

2. For $y = c_1 \cos \omega x$:

- $y' = -c_1 \omega \sin \omega x$
- $y'' = -c_1 \omega^2 \cos \omega x$
- $y'' + \omega^2 y = -c_1 \omega^2 \cos \omega x + \omega^2 \cdot c_1 \cos \omega x = 0$

Solution:

5. For $y = c_1 \cos \omega x + c_2 \sin \omega x$:

- $y' = -c_1\omega \sin \omega x + c_2\omega \cos \omega x$
- $y'' = -c_1\omega^2 \cos \omega x - c_2\omega^2 \sin \omega x$
- $y'' + \omega^2 y = -\omega^2(c_1 \cos \omega x + c_2 \sin \omega x) + \omega^2(c_1 \cos \omega x + c_2 \sin \omega x) = 0$

6.

$$y = A \sin(\omega x + B)$$

$$y' = A\omega \cos(\omega x + B)$$

$$y'' = -A\omega^2 \sin(\omega x + B)$$

$$y'' + \omega^2 y = -A\omega^2 \sin(\omega x + B) + \omega^2 \cdot A \sin(\omega x + B) = 0$$

- Verify that each of the following functions is a solution to the corresponding differential equation:

1. $y = \frac{\sin x}{x}$, $xy' + y = \cos x$;

2. $y = 2 + c\sqrt{1 - x^2}$, $(1 - x^2)y' + xy = 2x$ (where c is an arbitrary constant);

3. $y = ce^x$, $y'' - 2y' + y = 0$ (where c is an arbitrary constant);

4. $y = e^x$, $y'e^{-x} + y^2 - 2ye^x = 1 - e^{2x}$;

5. $y = \sin x$, $y' + y^2 - 2y\sin x + \sin^2 x - \cos x = 0$;

6. $y = -\frac{1}{x}$, $x^2y' = x^2y^2 + xy + 1$;

7. $y = x^2 + 1$, $y' = y^2 - (x^2 + 1)y + 2x$;

8. $y = -\frac{g(x)}{f(x)}$, $y' = \frac{f'(x)}{g(x)}y^2 - \frac{g'(x)}{f(x)}$.

Solution:

3. $y = ce^x$, $y'' - 2y' + y = 0$ (c is an arbitrary constant)

- Compute $y' = ce^x$, $y'' = ce^x$
- Substitute: $ce^x - 2ce^x + ce^x = 0$. **Verified**

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5. $y = \sin x, y' + y^2 - 2y \sin x + \sin^2 x - \cos x = 0$

- Compute $y' = \cos x$
- Substitute: $\cos x + \sin^2 x - 2 \sin x \cdot \sin x + \sin^2 x - \cos x = 0$. **Verified**

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6. $y = -\frac{1}{x}$, $x^2y' = x^2y^2 + xy + 1$

- Compute $y' = \frac{1}{x^2}$
- Substitute: $x^2 \cdot \frac{1}{x^2} = 1$, and $x^2 \cdot \left(-\frac{1}{x}\right)^2 + x \cdot \left(-\frac{1}{x}\right) + 1 = 1 - 1 + 1 = 1$. **Verified**

- Given the first-order differential equation $\frac{dy}{dx} = 2x$

1. Find its general solution
2. Find the particular solution passing through the point (1,4)
3. Find the solution that is tangent to the line $y = 2x + 3$
4. Find the solution satisfying the condition $\int_0^1 y\, dx = 2$
5. Sketch the graphs of the solutions in 2, 3, and 4

Solution: For differential equation $\frac{dy}{dx} = 2x$,

1. General solution $\xrightarrow{\text{Integration}}$ $y = \int 2x\,dx = x^2 + c_1$ $\xrightarrow{\text{General solution}}$ $y = x^2 + c_1$

2. Particular solution passing through the point $(1,4)$ $\rightarrow$ $4 = 1 + c_1$ $\rightarrow$ $c_1 = 3$ $\rightarrow$ $y = x^2 + 3$

3. The solution that is tangent to the line $y = 2x + 3$ $\rightarrow$ slope $= 2$ $\rightarrow$ from the DE $y' = 2x = 2$ $\rightarrow x = 1$

(it happens at $x = 1$)

$\rightarrow y = 2(1) + 3 = 5$, solution that passes through point $(1,5)$ $\rightarrow 5 = 1 + c_1$ $\rightarrow$ $c_1 = 4$ $\rightarrow y = x^2 + 4$

4. The solution satisfying $\int_0^1 y\,dx = 2$ $\rightarrow$ $\int_0^1 (x^2 + c_1)\,dx = 2$

$\rightarrow \left(\frac{1}{3}(1)^3 + c_1(1)\right) - 0 = 2$ $\rightarrow$ $c_1 = 2 - \frac{1}{3} = \frac{5}{3}$ $\rightarrow$ $y = x^2 + \frac{5}{3}$

- Find the common solution of the following two differential equations

$$y' = y^2 + 2x - x^4,$$

$$y' = 2x + x^2 + x^4 - y - y^2.$$

Solution:

$$y' = y^2 + 2x - x^4,$$

$$y' = 2x + x^2 + x^4 - y - y^2.$$

Set them equal $\longrightarrow$ $y^2 + 2x - x^4 = 2x + x^2 + x^4 - y - y^2$

After rearrangement $2y^2 + y - 2x^4 - x^2 = 0$ $\longrightarrow$ $2\left[\left(y+\tfrac{1}{2}\right)^2 - \tfrac{1}{4}\right] - 2\left[\left(x^2+\tfrac{1}{2}\right)^2 - \tfrac{1}{4}\right] = 0$

$$\longrightarrow \left(y+\frac{1}{2}\right)^2 = \left(x^2+\frac{1}{2}\right)^2$$

$$\longrightarrow y+\frac{1}{2} = x^2+\frac{1}{2} \quad \text{or} \quad y+\frac{1}{2} = -\left(x^2+\frac{1}{2}\right)$$

$$\longrightarrow y = x^2 \quad \text{or} \quad y = -x^2 - 1$$

- Find the straight line solutions of the differential equation $y' + xy'^2 - y = 0.$

(Find those solutions of the given differential equation which are $y = ax + b$)

Solution:

The solutions in the form of $y = ax + b$ of the differential equation $y' + xy'^2 - y = 0$.

$$y' = a \rightarrow a + xa^2 - ax - b = 0 \rightarrow x(a^2 - a) + a - b = 0 \rightarrow a^2 - a = 0 \;\; \& \;\; a - b = 0$$

$$a = 0 \;\; or \;\; a = 1$$

$$b = 0 \qquad b = 1$$

$$y = 0 \qquad y = x + 1$$

- Prove that the curve that is symmetric about the origin (0,0) with a solution curve of the differential equation $4x^2y'^2 - y^2 = xy^3$ is also a solution curve of this differential equation.

In other words,

if $y(x)$ is a solution curve of the given DE, then $-y(-x)$ is also a solution curve

or

if (x, y) is on the solution curve of the given DE, then $(-\mathrm{x}, -y)$ is also on the solution curve of this DE

Solution:

In fact, we want to show if the solution curve includes (x, y) then it also includes $(-x, -y)$

Assume $X = -x, \quad \rightarrow \quad \frac{dX}{dx} = -1,$

$Y = -y \quad \rightarrow \quad \frac{dY}{dx} = -\frac{dy}{dx}$

$$\Rightarrow \quad y' = \frac{dy}{dx} = -\frac{dY}{dx} = -\frac{dY}{dX}\frac{dX}{dx} = Y'$$

$$4x^2y'^2 - y^2 = xy^3 \xrightarrow[\substack{x = -X, \\ y = -Y, \\ y' = Y'}]{\text{Replacement}} 4(-X)^2(Y')^2 - (-Y)^2 = (-X)(-Y)^3$$

$$\rightarrow \quad 4X^2Y'^2 - Y^2 = XY^3$$

✓ **Verified**

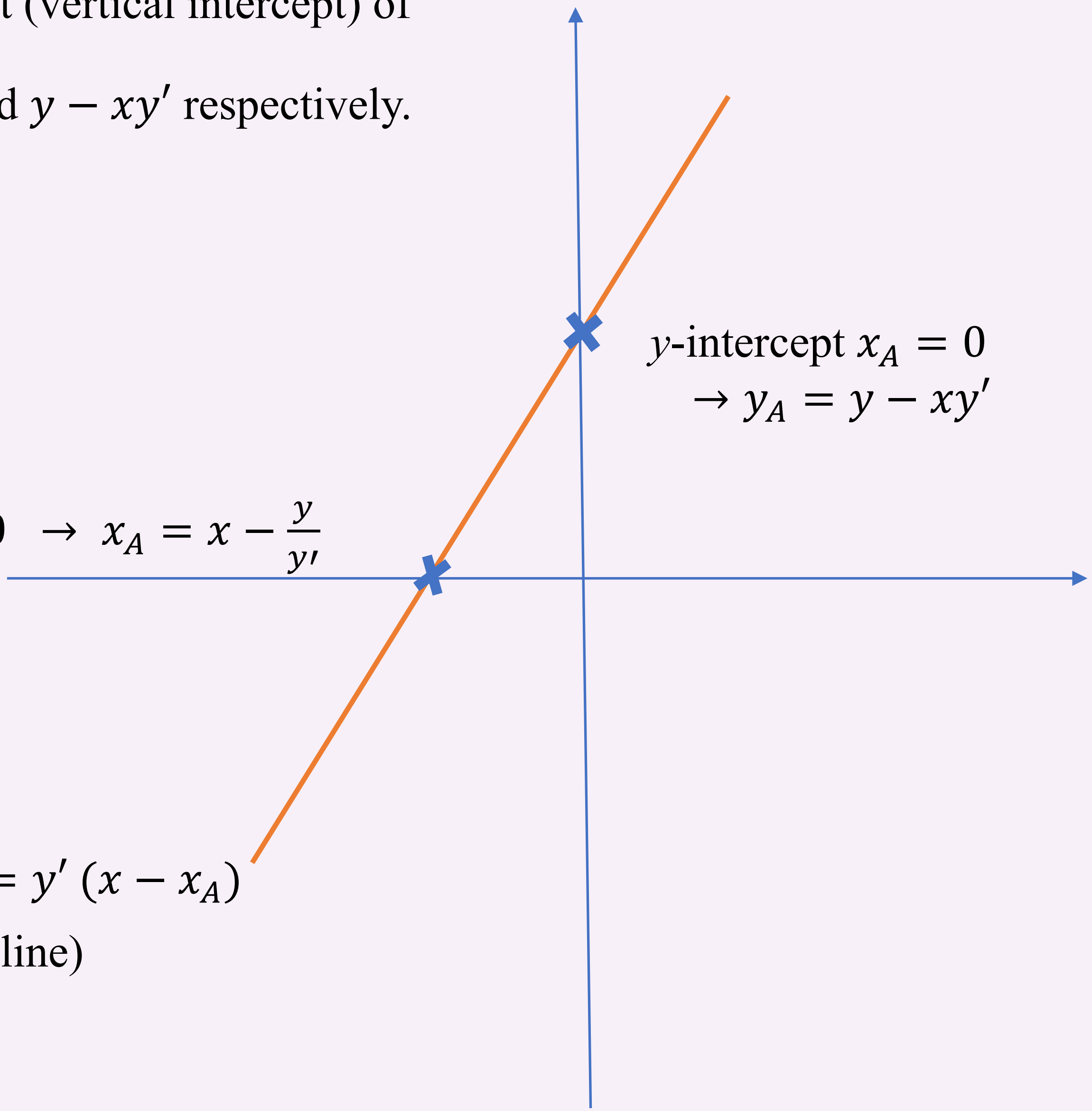
For each given property, find the differential equations which satisfies the conditions

(no need to solve the obtained differential equations)

(Homework #2)

1. The angle between the tangent at any point on the curve and the radial line (line connecting the origin to this point) at that point is α
2. The segment of the tangent at any point on the curve between the two axes is equal to a fixed length l
3. The area of the triangle enclosed by the tangent at any point on the curve and the two axes is equal to the constant a^2
4. The segment of the tangent at any point on the curve between the two axes is bisected by the point of tangency
5. The y-intercept of the tangent at any point on the curve is equal to the square of x-coordinate of that point of tangency
6. The y-intercept of the tangent at any point on the curve is the arithmetic mean of the x-coordinate and y-coordinate of that point of tangency
7. The slope of the tangent at any point on the curve is proportional to the x-coordinate of that point of tangency

Hint: The x-intercept (horizontal intercept) and y-intercept (vertical intercept) of the tangent passing through the point (x, y) are $x - \frac{y}{y'}$ and $y - xy'$ respectively.



formula of the line $y - y_A = y' \, (x - x_A)$

(A could be any point on the line)

(Homework #2)

试建立分别具有下列性质的曲线所满足的微分方程：

1. 曲线上任一点的切线与该点的径向夹角为α；
2. 曲线上任一点的切线介于两坐标轴之间的部分等于定长l；
3. 曲线上任一点的切线与两坐标轴所围成的三角形的面积都等于常数a^2；
4. 曲线上任一点的切线介于两坐标轴间的部分被切点等分；
5. 曲线上任一点的切线的纵截距等于切点横坐标的平方；
6. 曲线上任一点的切线的纵截距是切点的横坐标和纵坐标的等差中项；
7. 曲线上任一点的切线的斜率与切点的横坐标成正比.

(提示：过点(x,y)的切线的横截距和纵截距分别为 $x-\frac{y}{y'}$ 和 $y-xy'$)

- This is the **Logistic differential equation**, a fundamental model in population dynamics and other fields.

$$\frac{\mathrm{d}N}{\mathrm{d}t} = r\left(1 - \frac{N}{N_m}\right)N, \quad N(t_0) = N_0, \quad N(t) \geq 0$$

where:

- $\frac{\mathrm{d}N}{\mathrm{d}t}$ is the rate of change of N with respect to t,
- r is the intrinsic growth rate,
- N_m is the carrying capacity (maximum sustainable value of N),
- $N(t_0) = N_0$ is the initial condition,
- $N(t) \geq 0$ indicates N is non-negative.

Solve the differential equation and explain the solution.

Solution:

$$\frac{\mathrm{d}N}{\mathrm{d}t} = r\left(1 - \frac{N}{N_m}\right)N$$

$$\frac{\mathrm{d}N}{N\left(1 - \frac{N}{N_m}\right)} = r\mathrm{d}t$$

$$\int\left(\frac{1}{N} + \frac{1}{N_m - N}\right)\mathrm{d}N = \int r\mathrm{d}t$$

$$\ln|N| - \ln|N_m - N| = rt + C$$

$$\ln\left|\frac{N}{N_m - N}\right| = rt + C$$

$$\downarrow \quad e^{rt+C} = e^{rt}e^{C} = Ke^{rt}$$

C and K are parameters

General solution $$\frac{N}{N_m - N} = Ke^{rt}$$

Now we use the initial condition $N(t_0) = N_0$

Use the initial condition to find parameter K

$$N(t_0) = N_0 \longrightarrow \frac{N_0}{N_m - N_0} = Ke^{rt_0} \Longrightarrow K = \frac{N_0}{N_m - N_0}e^{-rt_0}$$

Substitute K in the solution

$$\longrightarrow \frac{N}{N_m - N} = \frac{N_0}{N_m - N_0}e^{r(t-t_0)}$$

all divided by $N_0e^{r(t-t_0)}$

$$= \frac{N_m}{\frac{N_m - N_0}{N_0e^{r(t-t_0)}} + \frac{N_0e^{r(t-t_0)}}{N_0e^{r(t-t_0)}}}$$

$$= \frac{N_m}{\frac{N_m - N_0}{N_0} \cdot e^{-r(t-t_0)} + 1}$$

Change it to the explicit form

$$N(t) = \frac{N_mN_0e^{r(t-t_0)}}{N_m - N_0 + N_0e^{r(t-t_0)}} \xrightarrow{\text{simplification}} \boxed{N(t) = \frac{N_m}{1 + \left(\frac{N_m}{N_0} - 1\right)e^{-r(t-t_0)}}}$$

As $t \to \infty$, $N(t) \to N_m$

It describes the **population size** (or population quantity) at a specific time t.

Remark

$$\boxed{\frac{\mathrm{d}y}{\mathrm{d}x} = g\left(\frac{y}{x}\right)}$$

or

$$y' = g\left(\frac{y}{x}\right)$$

$$\Rightarrow$$

$$y = ux$$

$$\frac{dy}{dx} = x\frac{du}{dx} + u$$

$$\boxed{\frac{\mathrm{d}y}{\mathrm{d}x} = f(ax + by + c) \quad (a \neq 0, b \neq 0)}$$

or

$$y' = f(ax + by + c)$$

$$\Rightarrow$$

$$u = ax + by + c$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = a + b\frac{\mathrm{d}y}{\mathrm{d}x}$$

- Solve $x\frac{dy}{dx} + 2\sqrt{xy} = y$ $(x < 0)$

Solution: $x\frac{dy}{dx} + 2\sqrt{xy} = y \qquad (x < 0)$

Rearrangement

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{y-2\sqrt{xy}}{x} = \frac{y}{x} - 2\sqrt{\frac{y}{x}}$$

Substitution

$y = ux$

$\frac{dy}{dx} = x\frac{du}{dx} + u$

$\Rightarrow \quad x\frac{\mathrm{d}u}{\mathrm{d}x} + u = u - 2\sqrt{u}$

$\Rightarrow \quad \frac{\mathrm{d}u}{2\sqrt{u}} = -\frac{\mathrm{d}x}{x}$

$\Rightarrow \quad \sqrt{u} = -\ln|x| + C$

$(x < 0) \Rightarrow \quad \sqrt{\frac{y}{x}} = -\ln(-x) + C$

$\Rightarrow \quad \frac{y}{x} = (\ln(-x) + C)^2 \implies y = x(\ln(-x) + C)^2$

where $\ln(-x) + C > 0$

- Solve $\frac{\mathrm{d}y}{\mathrm{d}x} = (x+y)^2$

Solution:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = (x+y)^2$$

Define $u = x + y,\ \longrightarrow\ \frac{du}{dx} = 1 + \frac{dy}{dx}$

Substitute $\frac{du}{dx} - 1 = u^2 \longrightarrow \frac{du}{u^2+1} = dx$

$$\longrightarrow \arctan u = x + c \longrightarrow u = \tan(x+c)$$

$$\longrightarrow x + y = \tan(x+c)$$

$$\longrightarrow y = \tan(x+c) - x$$

Remark

$$\boxed{\frac{\mathrm{d}y}{\mathrm{d}x} = f\left(\frac{a_1x+b_1y+c_1}{a_2x+b_2y+c_2}\right)}$$

Case 1: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = k$

Let $u = a_2x + b_2y$, then the equation becomes separable.

Case 2: $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

Solve the system $\begin{cases} a_1x + b_1y + c_1 = 0 \\ a_2x + b_2y + c_2 = 0 \end{cases}$ to find the intersection point (α, β).

Define $\begin{cases} X = x - \alpha \\ Y = y - \beta \end{cases}$ $\Rightarrow$ $\frac{\mathrm{d}Y}{\mathrm{d}X} = f\left(\frac{a_1X+b_1Y}{a_2X+b_2Y}\right)$

Now define $u = \frac{Y}{X}$

- Solve $\dfrac{dy}{dx} = \dfrac{x-y+1}{x+y-3}$

Solution:

$$\frac{dy}{dx} = \frac{x-y+1}{x+y-3}$$

Solve $\begin{cases} x-y+1=0 \\ x+y-3=0 \end{cases} \xrightarrow{x=1,\,y=2}$ Define $\begin{cases} X = x-1 \\ Y = y-2 \end{cases}$

Substitute $\frac{dY}{dX} = \frac{(X+1)-(Y+2)+1}{(X+1)+(Y+2)-3} = \frac{X-Y}{X+Y}$ Define $u = \frac{Y}{X} \longrightarrow \frac{dY}{dX} = X\frac{du}{dX} + u$

Substitute $X\frac{du}{dX} + u = \frac{1-u}{1+u} \longrightarrow \frac{1+u}{1-2u-u^2}du = \frac{dX}{X} \longrightarrow -\frac{1}{2}\ln|1-2u-u^2| = \ln|X| + c$

rewrite as $\frac{1+u}{2-(u+1)^2}$, then $\int \frac{1+u}{1-2u-u^2}du = -\frac{1}{2}\ln|1-2u-u^2| + c_1$

$$u = \frac{Y}{X}$$

$$\begin{cases} X = x - 1 \\ Y = y - 2 \end{cases}$$

$$-\frac{1}{2}\ln\left|1 - 2u - u^2\right| = \ln|X| + c$$

$$-\frac{1}{2}\ln\left|1 - 2\cdot\frac{y-2}{x-1} - \left(\frac{y-2}{x-1}\right)^2\right| = \ln|x-1| + c$$

Continue to simplify the solution

$$\ln\left|1 - 2\cdot\frac{y-2}{x-1} - \left(\frac{y-2}{x-1}\right)^2\right| = -2\ln|x-1| - 2c$$

$$\left|1 - 2\cdot\frac{y-2}{x-1} - \left(\frac{y-2}{x-1}\right)^2\right| = e^{-2\ln|x-1|-2c} = e^{-2c}\cdot|x-1|^{-2}$$

$$1 - 2\cdot\frac{y-2}{x-1} - \left(\frac{y-2}{x-1}\right)^2 = \frac{c_1}{(x-1)^2} \quad (c_1 = e^{-2c})$$

$$\boxed{(x-1)^2 - 2(x-1)(y-2) - (y-2)^2 = c_1}$$

- Solve $\frac{dy}{dx} = \frac{x - y + 1}{2x - 2y - 3}$

Solution:

$$\frac{dy}{dx} = \frac{x-y+1}{2x-2y-3}$$

Recall

Case 1: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = k$

Let $u = a_2x + b_2y$, then the equation becomes separable.

Case 2: $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

Solve the system $\begin{cases} a_1x + b_1y + c_1 = 0 \\ a_2x + b_2y + c_2 = 0 \end{cases}$ to find the intersection point (α, β).

Define $u = x - y, \longrightarrow \frac{du}{dx} = 1 - \frac{dy}{dx}$

Substitute $1 - \frac{du}{dx} = \frac{u+1}{2u-3} \longrightarrow \frac{du}{dx} = 1 - \frac{u+1}{2u-3} = \frac{2u-3-u-1}{2u-3} = \frac{u-4}{2u-3}$

Separate them $\left(2 + \frac{5}{u-4}\right) du = dx \longrightarrow 2u + 5\ln|u-4| = x + c$

$\longrightarrow 2(x-y) + 5\ln|x-y-4| = x + c$

$\longrightarrow x - 2y + 5\ln|x-y-4| = c$

RC Circuit Voltage Problem

- Based on Kirchhoff's Second Law and the capacitor current relation $I = C\frac{du_C}{dt}$, the equation for the charging process is:

$$RC\frac{du_C}{dt} + u_C = E$$

which describes how fast u_C changes (how quickly rises or falls), where

R: Resistance

C: Capacitance ($R \times C$ determines how fast u_C changes)

E: Power supply voltage

If capacitor is uncharged at first (initial condition is $u_C(0) = 0$), find the formula for u_C

Symbols and notations:

u_C : Capacitor voltage (the voltage across the capacitor)

t: Time (independent variable)

R: Resistance. This is the fixed resistance value of the resistor in the circuit (a built-in parameter of the circuit).

C: Capacitance. This is the fixed capacitance value of the capacitor in the circuit (also a built-in parameter of the circuit).

E: Power supply voltage. This is the fixed voltage of the battery or power source connected to the circuit.

$u_C(0)$: Initial capacitor voltage. This means the voltage of the capacitor at the starting moment (when time t=0). For example, $u_C(0)$ means the capacitor has no charge (and thus no voltage) at the very beginning.

Remark

The differential equation for the charging process: $RC\frac{du_C}{dt} + u_C = E$

The differential equation for the discharging process: $RC\frac{du_C}{dt} + u_C = 0$

Solution:

$$RC\frac{du_C}{dt}+u_C=E$$

Separate them $\dfrac{du_C}{u_C-E}=-\dfrac{dt}{RC} \longrightarrow \ln|u_C-E|=-\dfrac{t}{RC}+c_1 \xrightarrow{c_2=\pm e^{c_1}} u_C-E=c_2e^{-\frac{t}{RC}}$

$$\longrightarrow u_C=E+c_2e^{-\frac{t}{RC}}$$

$$u_C(0)=0 \longrightarrow 0=E+c_2 \longrightarrow c_2=-E \longrightarrow \boxed{u_C=E\left(1-e^{-\frac{t}{RC}}\right)}$$

$\tau=RC$

is the time constant

$t\to+\infty,\ u_C\to E.$

$t=0$: $u_C=0$

$t=\tau$: $u_C\approx 0.632E$

$t=2\tau$: $u_C\approx 0.865E$

$t=4\tau$: $u_C\approx 0.982E$

$t=5\tau$: $u_C\approx 0.993E$

- In the obtained RC circuit,

let E=10V, R=100Ω, C=0.01F,

and initially there is no charge on capacitor C

1. After switch S is turned to "1", how long does it take for the voltage u_C on capacitor C to reach 5V?

2. After switch S is turned to "1" for a very long time (e.g., 1 minute), switch S is suddenly turned to "2".

Try to find the variation law of u_C, and how long does it take for u_C to reach 5V?

Note:

switch S is turned to "1”: the circuit is initiating the charging process

switch S is turned to "2”: the circuit is initiating the discharging process

Solution:

1- Charging Phase (Switch at "1")

Differential equation is

$$RC\frac{du_C}{dt} + u_C = E$$

Since $u_C(0) = 0$, the solution is

$$u_C = E\left(1 - e^{-\frac{t}{RC}}\right)$$

Substitute

E=10V,

R=100Ω,

C=0.01F

Now we are looking for t when $u_C = 5V$:

$$5 = 10\left(1 - e^{-t}\right) \implies 1 - e^{-t} = 0.5 \implies e^{-t} = 0.5 \implies t = \ln 2 \approx 0.693\,\mathrm{s}$$

2- Discharging Phase (Switch at "2" after long time)

Differential equation of discharging is

$$RC\frac{du_C}{dt} + u_C = 0$$

First we have to find the solution with initial condition $u_C(0) = 10V$

Remember that for charginging process, $u_C = E\left(1 - e^{-\frac{t}{RC}}\right)$, so

$t \to \infty, u_C \to E$

which means after long-time charging u_C approaches E (=10V in this question). So, the initial condition for discharging is $u_C(0) = 10V$

$$RC\frac{du_C}{dt} + u_C = 0 \xrightarrow{\text{Separate them}} \frac{du_C}{u_C} = -\frac{1}{RC}dt$$

$$\xrightarrow{\text{Integration}} \ln|u_C| = -\frac{t}{RC} + c_1$$

$$u_C(t) = c_2 e^{-\frac{t}{RC}} \xrightarrow{u_C(0) = 10} u_C(t) = 10e^{-\frac{t}{RC}}$$

$$\xrightarrow[E=10\text{V},\ R=100\Omega,\ C=0.01\text{F}]{\text{Substitute}} 5 = 10e^{-t} \implies e^{-t} = 0.5 \implies t = \ln 2 \approx 0.693\,\text{s}$$

Exercises

1. $\dfrac{dy}{dx} = 2xy,\ y(0) = 1$

2. $y^2 \mathrm{d}x + (x+1)\mathrm{d}y = 0,\ y(0) = 1$

3. $\dfrac{dy}{dx} = \dfrac{1+y^2}{xy + x^3 y}$

4. $(1+x)y\mathrm{d}x + (1-y)x\mathrm{d}y = 0$

5. $(y+x)\mathrm{d}y + (x-y)\mathrm{d}x = 0$

6. $x\dfrac{dy}{dx} - y + \sqrt{x^2 - y^2} = 0$

7. $\tan y\, dx - \cot x\, dy = 0$

8. $\dfrac{dy}{dx} + \dfrac{\mathrm{e}^{y^2+3x}}{y} = 0$

9. $x(\ln x - \ln y)\, dy - y\, dx = 0$

10. $\dfrac{dy}{dx} = \mathrm{e}^{x-y}$

For solving 5, 6, 9 substitute $u = \frac{y}{x}$

Exercises (substitution)

1. $\dfrac{dy}{dx} = \dfrac{1}{(x+y)^2}$

2. $\dfrac{dy}{dx} = \dfrac{2x-y+1}{x-2y+1}$

3. $\dfrac{dy}{dx} = \dfrac{x-y+5}{x-y-2}$

$u = x + 4y$

4. $\dfrac{dy}{dx} = (x+1)^2 + (4y+1)^2 + 8xy + 1$

$u = \dfrac{y^3}{x}$

5. $\dfrac{dy}{dx} = \dfrac{y^6 - 2x^2}{2xy^5 + x^2y^2}$

$u = \dfrac{y}{x}$

6. $\dfrac{dy}{dx} = \dfrac{2x^3 + 3xy^2 + x}{3x^2y + 2y^3 - y}$

Exercises (substitution)

$$u = xy$$

1. $y(1+x^2y^2)\mathrm{d}x = x\mathrm{d}y$

2. $\dfrac{x}{y}\dfrac{dy}{dx} = \dfrac{2+x^2y^2}{2-x^2y^2}$

- $f(x)$ is given; find the general expression of the function $f(x)$ if

$$f(x) \int_0^x f(t)\,\mathrm{d}t = 1 \ (x \neq 0),$$

Solution:

Define $F(x) = \int_0^x f(t)\,\mathrm{d}t$

$\longrightarrow$ $F'(x) = f(x)$

It is given that $f(x)F(x) = 1 \implies F'(x)F(x) = 1$

$\int F(x)F'(x)\,\mathrm{d}x = \int 1\,\mathrm{d}x \implies \frac{1}{2}F(x)^2 = x + c \rightarrow F(x) = \pm\sqrt{2x + 2C}$

Since, $F(x)F'(x) = 1 \implies F'(x) = \frac{1}{F(x)}$

$\downarrow$

Also, $f(x) = F'(x) = \frac{\pm 1}{\sqrt{2x+2c}}$

$\downarrow$

$f(x) = \frac{c}{\sqrt{x}}$

- Find the function $x(t)$ with the property $x(t+s)=\dfrac{x(t)+x(s)}{1-x(t)x(s)}$ given that $x'(0)$ exists.

Hint: you may find a differential equation first

Solution:

Step 1: Determine $x(0)$

Substitute $t = s = 0$

$$\frac{2x(0)}{1-x^2(0)} = x(0) \implies 2x(0) = x(0)\left[1-x^2(0)\right] \implies x(0)\overset{\to 0}{\left[2-1-x^2(0)\right]} = 0 \longrightarrow x(0) = 0$$

Step 2: Derive the differential equation

According to the given property $x(t+h) = \dfrac{x(t)+x(h)}{1-x(t)x(h)}$

According to the definition of derivative $x'(t) = \lim_{h\to 0}\frac{x(t+h)-x(t)}{h}$

$$x'(t) = \lim_{h\to 0}\frac{\dfrac{x(t)+x(h)}{1-x(t)x(h)} - x(t)}{h} = \lim_{h\to 0}\frac{x(h)\left[1+x^2(t)\right]}{h\left[1-x(t)x(h)\right]} = \lim_{h\to 0}\left(\frac{x(h)}{h}\cdot\frac{1+x^2(t)}{1-x(t)x(h)}\right)$$

$$\longrightarrow x'(t) = \left(1+x^2(t)\right)\lim_{h\to 0}\frac{x(h)}{h} = x'(0)\left(1+x^2(t)\right) \xrightarrow{c\,=\,x'(0)} x'(t) = c\left(1+x^2(t)\right)$$

Step 3: Solve the differential equation

$$\int \frac{dx}{1+x^2} = \int c\,dt \implies \arctan(x) = ct + c_1$$

$x(0) = 0$

$$\longrightarrow \arctan(0) = 0 + c_1 \implies c_1 = 0 \longrightarrow \arctan(x) = ct$$

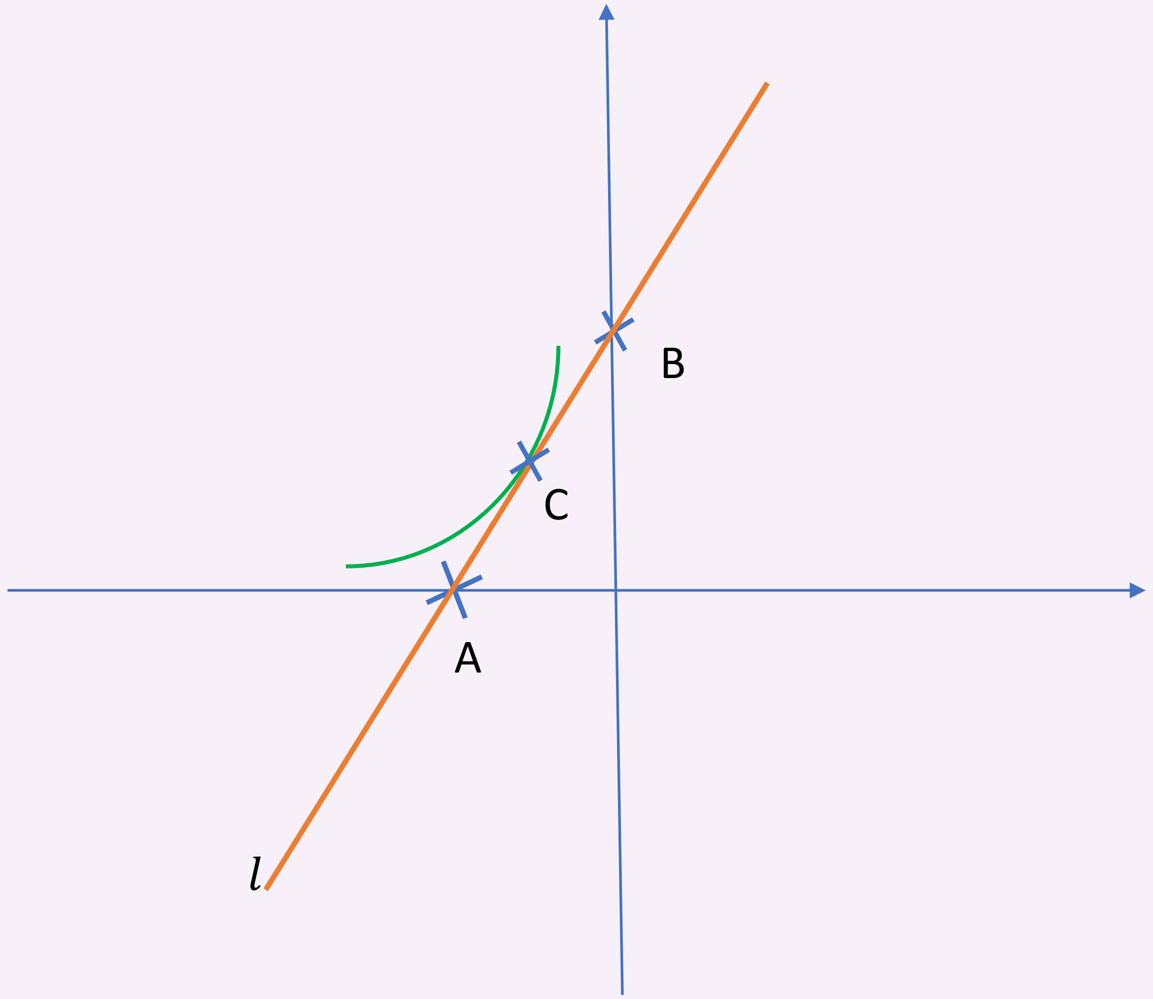
$$\longrightarrow x(t) = \tan(ct)$$

where $c = x'(0)$

- Find a curve such that the segment of its tangent line between the coordinate axes is divided into two equal parts by the point of tangency.

Hint:

For an arbitrary point C on this curve (the green curve in the figure), the tangency point (point C) is exactly in the middle of A and B. Find the curve!



Solution:

Step 1: Define the Curve and Tangent Line

- Assume $y = y(x)$ is that curve. Let ℓ be the tangent line at point $C(x_C, y_C)$.
- The slope of ℓ at C is $y'_C = \left.\frac{dy}{dx}\right|_{(x_C,y_C)}$, so the tangent line equation is: $y - y_C = y'_C(x - x_C)$

Step 2: Find points A and B

- Point A (Y-intercept, $x_A = 0$):

$$y_A - y_C = y'_C(0 - x_C) \implies y_A = -y'_C x_C + y_C$$

- Point B (X-intercept, $y_B = 0$):

$$0 - y_C = y'_C(x_B - x_C) \implies x_B = x_C - \frac{y_C}{y'_C}$$

Step 3: C is the midpoint of A and B

$$x_C = \frac{x_A+x_B}{2} = \frac{0+x_B}{2} \implies 2x_C = x_B \implies 2x_C = x_C - \frac{y_C}{y'_C} \implies x_C = -\frac{y_C}{y'_C}$$

$$y_C = \frac{y_A+y_B}{2} = \frac{y_A+0}{2} \implies 2y_C = y_A \implies 2y_C = -y'_C x_C + y_C \implies y_C = -y'_C x_C$$

$$\Rightarrow \quad y' = -\frac{y}{x}$$

The solution of this equation gives the curve with the required property

Step 4: Solve the Differential Equation

$$y' = -\frac{y}{x} \longrightarrow \frac{dy}{y} = -\frac{dx}{x} \longrightarrow \ln|y| = -\ln|x| + c_1$$

$$\longrightarrow y = \frac{c_2}{x} \qquad (c_2 = e^{c_1})$$

$$\Rightarrow \quad \boxed{xy = c}$$